Deep Learning in Finance

Assignment 1 — Research Paper Presentation and Implementation

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Overview

Your first assignment is a **group presentation of a research paper**, delivered in class, together with a **GitHub repository that implements the paper or a well-chosen segment of it**. This assignment is worth **40%** of your course grade.

The point of the exercise is not to summarize a paper. It is to get to the stage where you could defend it: to know what the authors actually did, why the architecture they chose was the right one for the problem, what the paper establishes and what it merely asserts, and — above all — what happens when you try to run it yourself. The implementation is where most of the learning happens, and it is weighted accordingly.

Groups and Schedule

The class splits into **five groups**. Presentations run from **Week 3 through Week 7**, one group per session, at the start of class. Each group is assigned one of the course's seven topics; the topic is lectured *before* the presentation, so you present on material the room has already seen.

Group	**Presents**	**Topic**	**Lectured**
Group 1	Week 3	Theoretical Foundations and the Virtue of Complexity	Week 1
Group 2	Week 4	Transformers	Week 2
Group 3	Week 5	Neural ODEs and Neural Point Processes	Week 3
Group 4	Week 6	Generative Adversarial Networks	Week 4
Group 5	Week 7	Diffusion Models	Week 5

Two topics are deliberately left uncovered. **Graph Neural Networks** (Week 6) and **Bayesian Deep Networks for Factor Modeling** (Week 7) are lectured too late in the term to leave a group a reasonable gap between the lecture and its presentation slot. Both are lectured in the normal way; they simply carry no presentation.

Group size depends on enrollment — expect three to five students per group. **Every member presents.** A group that sends one person to the front while the rest sit down will be marked accordingly.

Choosing a Paper

You have two routes:

1. **Pick from the list** in the final section of this document. The PDFs are in `Assignments/papers/` in the course repository, with the exception noted there.
2. **Propose your own.** Any paper is fair game provided it is a genuine research paper (peer-reviewed, or a serious preprint), it sits squarely within your group's topic, and **I approve it**. Email me the citation and a two-or-three sentence case for it. I am happy to approve papers that are not on my list — a paper you are actually curious about will produce a better presentation than one you were handed. What I will turn down is a paper with no implementable core, a survey, or something so far from the topic that the rest of the class gains nothing.

No two groups may present the same paper. Choices are first-come, first-served, so claim yours early.

Deadlines for selection

- **End of Week 1:** group rosters finalized and emailed to me.
- **End of Week 2:** paper choice submitted (or proposal approved). Group 1 should settle both as early in Week 1 as possible, since they present in Week 3.

Deliverable 1 — The Presentation (10–15 minutes)

Ten to fifteen minutes is short. It is not enough time to walk through a paper section by section, and you should not try. Decide what the paper's one real idea is and spend your time there. A rough shape that works:

- **The problem (1–2 min).** What financial question is at stake, and why do the standard tools fail at it? Motivate it in the language of finance, not of machine learning.
- **The method (4–6 min).** The architecture and the objective. Write down the loss. Be precise about what is learned, what is fixed, and what the inputs and outputs actually are. If the paper's contribution is a modification of a standard architecture, make the modification visible.
- **The results (2–3 min).** The one or two tables or figures that carry the paper's claim. What is the baseline, and is it a fair one?

- **Your implementation (3–4 min).** What you built, what you reproduced, what you could not, and what you learned from the gap. This section is often the most interesting part of the talk — do not let it get squeezed out.
- **Your critique (1–2 min).** Where is the paper vulnerable? Look-ahead bias, transaction costs, a suspiciously favorable sample period, a baseline chosen to lose, hyperparameters tuned on test data. Most papers have a soft spot. Finding it is a skill this course is trying to teach you.

Expect questions, from me and from the class. Budget for them: aim your prepared content at the lower end of the window. Slides in any format; send them to me **24 hours before** your session.

Deliverable 2 — The GitHub Repository

Submit a link to a GitHub repository (public, or private with me added as a collaborator) that implements the paper or a segment of it.

Scope: depth over breadth

You are not expected to reproduce a whole paper. A faithful implementation of one mechanism, evaluated honestly on a small universe, is worth far more than a sprawling half-working attempt at everything. Legitimate ways to cut scope:

- One asset class, or one market, instead of the paper's full cross-section.
- A shorter sample period, or a smaller universe (e.g. 50–100 names).
- A reduced model — fewer layers, smaller hidden dimension, fewer random features — as long as the mechanism under study survives the cut.
- Simulated data where the paper's proprietary data is unobtainable. If the paper uses TAQ or CRSP and you cannot get it, build a simulator whose stylized facts you can state, and say clearly what that substitution costs you.

State your scope decisions in the README. A clearly reasoned reduction is a sign of judgment, not a shortfall. What is *not* acceptable is cloning the authors' repository and running it — if an official implementation exists, you may read it and should cite it, but the code you submit must be code you wrote and understand.

What the repository must contain

- **A README** that states: the full citation of the paper; what you implemented and what you deliberately left out; how to install and run it; and what you found. Someone landing on the repo cold should understand the project in two minutes.

- **Reproducibility.** A `requirements.txt` or environment file, fixed random seeds, and a single documented entry point — a script or a driver notebook — that regenerates your headline result. I will try to run it.
- **Structure.** Reusable logic in modules; notebooks for narrative and figures. Not one 2,000-line notebook.
- **At least one figure or table** that either reproduces a result from the paper or is the honest analogue of one under your reduced scope, placed side by side with the paper's number where a comparison is possible.
- **Data handling.** Commit a small sample or a fetch script. Do not commit large vendor extracts.
- **A "what didn't work" section.** Required, not optional. The thing you could not get to train, the result that came out backwards, the number you cannot reconcile with the paper. Write it down. A repository reporting a failed reproduction with a clear account of the failure scores well; one quietly reporting success it did not achieve does not.
- **An AI usage note** (see below).

Due: repository link emailed 24 hours before your session, with a tagged commit (e.g. `v1.0`) marking what you are submitting. You are welcome to keep working on it afterwards — I grade the tag.

On Coding Agents and LLMs

Use them. Heavily. This is encouraged, not merely tolerated.

Coding agents — Claude Code, Codex, Cursor, and whatever has shipped by the time you read this — are now part of how quantitative research is actually done, and you are expected to be fluent with them. Use an agent to scaffold your repository, to port an equation into a working module, to write the backtest harness, to track down the bug that has eaten your afternoon. There is no credit here for doing by hand what a tool does better.

Likewise, **use LLMs to demystify the hard parts.** These are difficult papers, often assuming machinery you have not met. When a paper asserts an identity and moves on, ask. Ask for the derivation in smaller steps; ask for the toy two-dimensional version of the construction; ask why the authors chose this estimator over the obvious alternative; ask it to re-explain the same object three different ways until one lands. Used this way, an LLM is the patient co-reader that makes a hard paper tractable. I would rather you attempted an ambitious paper with heavy LLM support than a safe one unaided.

Three conditions, and they are firm:

1. **You own what you submit.** Anything in your repository or your slides, you can explain and defend on the spot. I will ask about specific lines. "The agent

wrote that" is not an answer, and it is the one failure mode that this assignment is designed to expose. The implementation deliverable exists precisely because understanding survives it and cut-and-paste does not.

2. **Verify everything.** Agents and LLMs produce confident, plausible, wrong output — fabricated citations, subtly incorrect derivations, backtests with look-ahead baked in. Check the math against the paper. Check that your train/test split is what you think it is. A hallucinated result presented as the paper's is a serious error, and "the model told me" does not reduce your responsibility for it.
3. **Disclose it.** Include a short `AI_USAGE.md` (or a README section) describing which tools you used and for what — a paragraph is plenty. This is not a penalty; nothing in your grade turns on how much you used them. It is professional practice, and it mirrors the disclosure in this repository's own README.
4. **Do not let an agent write your talk.** Slides generated wholesale read exactly as such. The synthesis — deciding what the paper's one real idea is, and what to cut — is the part of this assignment that is yours.

Grading

Component	Weight
Implementation — correctness, reproducibility, and honesty of the repository	40%
Technical command of the paper — the method explained precisely and correctly	25%
Presentation — structure, clarity, use of the time, and handling of questions	20%
Critical assessment — identifying the paper's real weaknesses	15%

As with the final project, **reproducing the paper's headline number is not required for a strong grade.** Rigor, a correct account of the method, and a clear-eyed explanation of why your results do or do not match carry more weight than a matching Sharpe ratio. Groups are graded jointly, with an individual adjustment where one member's contribution is plainly out of line with the rest.

Paper List

Group 1 — Theoretical Foundations and the Virtue of Complexity

- Bryan Kelly, Semyon Malamud, Kangying Zhou (2022) The Virtue of Complexity Everywhere, SSRN Working Paper 4166368. *Not bundled — download requires a free SSRN account.*

Group 2 — Transformers

- Elliot L. Epstein, Rose Wang, Jaewon Choi, Markus Pelger (2025) Attention Factors for Statistical Arbitrage.

Group 3 — Neural ODEs and Neural Point Processes

- Jingsui Li, Wei Zhu, Zhang Chen, Chao Pei (2023) Financial Time Series Prediction via Neural Ordinary Differential Equations Approach, CSIS-IAC.

Group 4 — Generative Adversarial Networks

- Adil Rengim Cetingoz, Charles-Albert Lehalle (2025) Synthetic Data for Portfolios: A Throw of the Dice Will Never Abolish Chance.

Group 5 — Diffusion Models

- Zhuohan Wang, Carmine Ventre (2024) A Financial Time Series Denoiser Based on Diffusion Models, ICAIF.

Questions, paper proposals, and scope negotiations — come early rather than late, particularly about scope:

`oualid.missaoui@baruch.cuny.edu` or during office hours.